HSRP-Enabled Huawei Access Node BAS-ACC-05 Site-Wide Routing Protocol Flap Resolution SOP

Node: BAS-ACC-05 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_3351

Description

The failure of HSRP-enabled Huawei Access Node BAS-ACC-05 Site-Wide Routing Protocol Flap is a complex phenomenon resulting from the interplay of BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This condition occurs due to the inherent dynamics of the routing protocols at play, specifically the rapid oscillation of routing information between the nodes. As the protocols strive to converge on a consistent routing table, they may temporarily lose synchronization, leading to a cascade of routing flaps. This event further exacerbates the instability, creating a self-reinforcing cycle that threatens the integrity of the network. Furthermore, the Huawei access node's configuration settings, which are optimized for efficient routing operations, may inadvertently contribute to this condition by amplifying the oscillating effects. As such, it is crucial to carefully examine the configuration and topology of the network to identify any potential misconfigurations or issues that may be contributing to this problem.

Pre-requisite Checks

- Verify that the Basavanagudi POP is accessible via remote hands
- Check the status of redundant links and ensure they are operational
- Gather relevant system logs and configuration files from the Huawei access node

Potential Root Causes

Cause	Probability
BGP route churn due to incorrect routing policy	High
OSPF adjacency loss due to misconfigured OSPF settings	Medium
ISIS LSP flooding due to hardware issues	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table, Expected Output: Display the BGP routing table and verify for any anomalies, Description: This command will display the BGP routing table and help identify any routing policy errors that may be causing the BGP route churn.

Command: display ospf routing-table, Expected Output: Display the OSPF routing table and verify for any adjacency loss, Description: This command will display the OSPF routing table and help identify any OSPF adjacency loss issues that may be contributing to the routing flaps.

Command: display isis routing-table, Expected Output: Display the ISIS routing table and verify for any LSP flooding, Description: This
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```
command will display the ISIS routing table and help identify any ISIS
LSP flooding issues that may be causing the routing flaps.
```

Resolution Steps

Step 1: Step: Modify the BGP routing policy to prevent route churn, Command: set routing-policy bgp import 10 permit as-path 10, Description: This command will modify the BGP routing policy to prevent route churn by allowing only the specified AS paths to be imported.

Step 2: Step: Reconfigure OSPF settings to prevent adjacency loss, Command: set ospf process 10 network 10.1.1.0/24 area 0.0.0.1, Description: This command will reconfigure the OSPF settings to prevent adjacency loss by specifying the correct network and area for the OSPF process.

Step 3: Step: Verify and adjust the ISIS LSP flooding settings, Command: display isis lsp-flood-status, Description: This command will display the ISIS LSP flooding status and help identify any issues that may be causing the LSP flooding.

Post-Check Verification

```
Command: display bgp routing-table, Expected Output: Verify that the
BGP routing table is stable and free of anomalies, Description: This
command will verify that the BGP routing table is stable and free of
anomalies after the resolution steps have been applied.

Command: display ospf routing-table, Expected Output: Verify that the
OSPF routing table is stable and free of adjacency loss, Description:
This command will verify that the OSPF routing table is stable and
free of adjacency loss after the resolution steps have been applied.

Command: display isis routing-table, Expected Output: Verify that the
ISIS routing table is stable and free of LSP flooding, Description:
This command will verify that the ISIS routing table is stable and
free of LSP flooding after the resolution steps have been applied.
```

Roll-back Procedures

- Step: Undo the BGP routing policy modifications, Command: undo routing-policy bgp import 10 permit as-path 10, Description: This command will undo the BGP routing policy modifications and restore the original settings.

- Step: Revert the OSPF settings to their original configuration, Command: undo set ospf process 10 network 10.1.1.0/24 area 0.0.0.1, Description: This command will revert the OSPF settings to their original configuration and restore the original network and area settings.

- Step: Verify and adjust the ISIS LSP flooding settings to their original configuration, Command: undo display isis lsp-flood-status, Description: This command will verify and adjust the ISIS LSP flooding settings to their original configuration and restore the original settings.